

When Gossip Meets Flames: A Gossip Learning Framework for Enhancing UAV Swarm Robustness in Forest Fire Detection

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TABLE I
ABBREVIATION TABLE

Abbreviation	Meaning
UAV	Unmanned Aerial Vehicle
FFD	Forest Fire Detection
UAV-OD	Unmanned Aerial Vehicle Object Detection
FL	Federated Learning
GL	Gossip Learning
GC	Gravitational Clustering
GB	Gossip Backup
GS	Gossipmonger Selection
RGLF	Robust Gossip Learning Framework
AI	Artificial Intelligence
ML	Machine Learning
D2D	Device-to-Device
GCN	Graph Convolutional Network
Adam	Adaptive Moment Estimation
PFL	Partial Federated Learning
RGLF-EX	RGLF under extreme risks of UAV incapacitation

Abstract—Deploying Unmanned Aerial Vehicle (UAV) swarm for Forest Fire Detection (FFD) in hostile environments poses critical dependability challenges to decentralized learning systems, where UAVs are vulnerable to damage from extreme heat, turbulent air currents, and dense smoke. Such damage can lead to data loss and compromise the effectiveness of FFD missions. To address these challenges, we propose a Robust Gossip Learning Framework (RGLF), a fault-tolerant and decentralized learning framework that enhances the robustness and adaptability of UAV swarm-based FFD missions. RGLF introduces a gravitational clustering method to organize UAVs into dispersed clusters, a gossip backup method for decentralized data redundancy, and an adaptive gossipmonger selection method that dynamically designates representative UAVs for model training, preventing mission failures due to the gossipmonger incapacitation. By eliminating reliance on a central coordinator, RGLF enables fault-tolerant model updating through decentralized gossip learning, ensuring continuous learning under the UAV incapacitation. Experimental results demonstrate that RGLF can significantly enhance the mission performance, even when some UAVs are incapacitated. By strengthening the UAV swarm robustness, RGLF provides a more dependable and scalable solution for UAV swarm-based FFD missions.

Index Terms—UAV swarm robustness, forest fire detection, decentralized gossip learning, fault-tolerant model updating.

I. INTRODUCTION

IN RECENT years, Unmanned Aerial Vehicles (UAVs) have demonstrated significant potential across various fields—including emergency communications [1], disaster response, military surveillance [2], agricultural monitoring [3], and Forest Fire Detection (FFD) [4]. When deployed as a swarm, UAVs gain enhanced intelligence, coordination, flexibility, survivability, and adaptability, enabling them to execute complex missions with superior performance compared with a single UAV [5]. For clarity, Table I summarizes the abbreviations used throughout this paper.

Received 1 January 2026; revised 2 April 2026; accepted 3 June 2026. Date of publication 8 June 2026; date of current version 1 September 2026. This work was supported in part by the National Natural Science Foundation of China under Grant 62272237 and Grant 61872191 and in part by the Postgraduate Research & Practice Innovation Program of Jiangsu Province under Grant KYCX25_1148. (Corresponding authors: Kejia Chen; Linfeng Liu.)

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Digital Object Identifier 10.1109/TDSC.2026.3701006

Driven by the global climate changes and human activities, forest fires have grown increasingly frequent and intense, posing severe threats to ecosystems and human communities [6], [7]. Timely and accurate FFD is therefore crucial for early warning systems and rapid emergency response. In particular, FFD focuses on detecting small fires at their incipient stages and plays a vital role in minimizing the ecological and infrastructural damage. Currently, UAV swarm has emerged as a highly promising solution for FFD [8], e.g., UAV swarm can efficiently patrol vast and remote forested regions while collecting critical fire detection data [9].

To perform FFD missions, the UAV swarm requires an Unmanned Aerial Vehicle Object Detection (UAV-OD) model [10]. However, the real-world deployment of UAV-OD model often faces some limitations in generalization capability, as hostile mission environments and evolving fire scenarios typically pose significant challenges to FFD missions [11]. Consequently, the fault-tolerant model updating are essential to enable the UAV swarm to effectively adapt to hostile mission environments and evolving fire behaviors.

The UAV swarm leverages real-time visual data (videos and images) captured during surveillance to update the parameters of UAV-OD model. Federated Learning (FL), a distributed learning framework, is well-suited for implementing the fault-tolerant

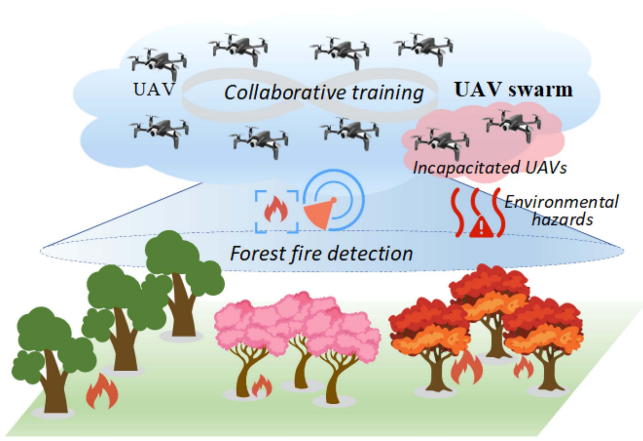


Fig. 1. UAV swarm in forest fire environments.

model updating, as it enables collaborative training of the UAV-OD model across the UAV swarm [12]. Moreover, as a decentralized variant of FL, Gossip Learning (GL) proves more efficient for UAV swarm in wireless resource-constrained scenarios [13], significantly reducing the training overhead while enhancing the adaptability of UAV swarm-based FFD missions.

However, some challenging environments inherent to forest fires inevitably expose UAV swarm to environmental hazards (e.g., extreme heat, turbulent air currents, and dense smoke), which could lead to the incapacitation of some UAVs, where the UAV incapacitation refers to the physical damage or complete communication failure that prevents UAVs from further participating in FFD missions. As illustrated in Fig. 1, such environments can result in the loss of data collected by incapacitated UAVs, which not only disrupts the fault-tolerant model updating but also ultimately impairs the performance of FFD missions.

Therefore, it is crucial to investigate the robustness of UAV swarm in FFD missions when some UAVs are incapacitated. A cluster is a group of UAVs organized to improve the coordination and communication efficiency. Clustering method helps manage the UAV swarm more effectively during FFD missions. Intuitively, while UAV clusters can be naturally formed based on the spatial proximity [14], this approach becomes vulnerable when fire-induced thermal updrafts affect an airspace, potentially destroying all UAVs in this airspace and causing the complete loss of local data of these incapacitated UAVs. To address this limitation, we propose: (i) a Gravitational Clustering (GC) method that organizes UAVs into geographically dispersed clusters; (ii) a Gossip Backup (GB) method that minimizes the data loss from incapacitated UAVs while avoiding the excessive training overhead of traditional centralized backup approaches; and (iii) a Gossipmonger Selection (GS) method to maintain the FFD mission continuity when one or more gossipmongers become incapacitated.

Furthermore, to address the limitations of conventional centralized or partially decentralized learning frameworks and enhance the robustness and optimize the training efficiency of UAV swarm in FFD missions, we specially design a GL protocol, where each UAV cluster elects a designated gossipmonger to

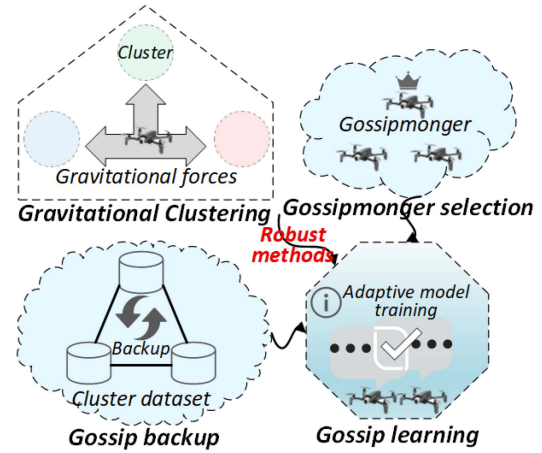


Fig. 2. Basic methodology of our proposed RGLF.

engage in collaborative training of the gossip model. Each gossipmonger aggregates the received gossip models (parameters) from peer gossipmongers, and performs the local training based on its dataset, then disseminates the updated gossip model parameters to peer gossipmongers. This decentralized paradigm enables iterative model training, parameter exchanges, and parameter aggregation among UAVs without any centralized coordination. Key advantages of our proposed framework include: significant reduction in training overhead during model training, enhanced adaptability of UAV swarm-based FFD missions through fault-tolerant model updating, and elimination of single-point failure risks inherent in centralized approaches. In this context, the designed GL protocol serves as a fault-tolerant and decentralized learning that maintains the continuity and robustness in UAV swarm-based FFD missions.

The UAV swarm is modeled as a mission-driven sensing swarm for FFD missions. The basic functionalities of individual UAVs, including video recording, communication, and local model training, are assumed to be available. The proposed Robust Gossip Learning Framework (RGLF) is designed to enable fault-tolerant and decentralized learning, thereby ensuring mission continuity under UAV incapacitations.

To provide an overview of the key considerations, the basic methodology of RGLF is illustrated in Fig. 2. The main contributions of this work are threefold:

- We present a RGLF that employs fault-tolerant model updating to improve model generalization, thereby ensuring fault-tolerant and adaptive FFD mission performance for UAV swarm in hostile forest fire environments.
- We specially develop three fault-tolerant methods to enhance the UAV swarm robustness against the UAV incapacitation: a gravitational clustering method, a gossip backup method, and a gossipmonger selection method.
- The proposed RGLF implements a decentralized gossip learning protocol that achieves adaptive model training while significantly reducing the training overhead.

The remainder of this paper is organized as follows: Section II briefly surveys some existing related studies. Section III provides

a system model and problem formulation for the robustness of UAV swarm in FFD missions. Section IV proposes Robust Gossip Learning Framework (RGLF). Section V covers some further analyses on RGLF, including complexity, model convergence. Simulation results for performance evaluation of RGLF are reported in Section VI. Finally, Section VII concludes the paper.

II. RELATED WORK

A. Distributed UAV Swarm

Distributed UAV swarm has been extensively investigated to provide solutions for various cooperative operations in the UAV swarm. In [15], a controlling framework is proposed to integrate the modes of formation flight and swarm deployment over both fixed and switching topologies, effectively solving the formation flight and obstacle avoidance problems. A hierarchical distributed framework for decision-making and collaborative control is developed in [16] to enable self-organized formation in the UAV swarm. [17] proposes an integrated framework for cooperative control of UAVs, combining distributed control strategies with human-swarm interaction mechanisms, exhibiting enhanced obstacle avoidance capabilities and robust disturbance rejection.

Communication-aware and sensing-driven collaboration has also been widely explored in the distributed UAV swarm. In [18], wideband spectrum sensing augmented technology with a Nyquist folding receiver is introduced to improve the utilization of spectrum for the distributed UAV swarm. A distributed UAV-enabled integrated sensing and communication system is proposed in [19] to simultaneously perform sensing and wireless communication, where multi-agent reinforcement learning is adopted to support autonomous decision-making in time-varying environments. In addition, [20] develops a novel framework for decentralized semantic communication and cooperative tracking control in the UAV swarm, which selectively transmits informative control data to significantly reduce communication overhead and energy consumption while maintaining accurate swarm coordination.

To address the accurate sensing and achieve energy-efficient communication, [21] investigates a symbiotic communication technology with super-resolution sensing to jointly enhance sensing accuracy and communication efficiency in low-altitude UAV swarm. From the perspective of secure communication, [22] proposes a privacy-preserving physical layer authentication framework for UAV-assisted communication systems, enabling reliable and secure UAV identity verification through carrier frequency offset and elliptic curve cryptography.

Learning-enabled distributed decision-making has further attracted increasing attention for the distributed UAV swarm. In [23], a distributed generative reinforcement learning approach is proposed for service function chain orchestration in highly dynamic UAV swarm networks, resulting in lower computational overhead and faster convergence. A decentralized and cooperative UAV swarm control approach based on symmetry-informed multi-agent reinforcement learning is developed in [24], where

a symmetry-informed graph neural network is designed to improve training efficiency for large swarms.

Despite these advances, most existing studies concentrate on swarm coordination and control, while limited attention has been paid to ensuring robustness and mission continuity in the event of UAV incapacitation.

B. UAV Swarm for Forest Fire Detection

UAV swarm has garnered significant attention for addressing distributed, computation-intensive missions such as FFD. Given the limited visual coverage and computational capacity of individual UAVs, collaborative approaches have been developed to enhance the efficiency and robustness of UAV swarm-based FFD missions. Through collaborative clustering, UAV swarm can effectively execute complex missions.

Early FFD via UAVs presents two major challenges: the typically small fire signatures and their susceptibility to environmental interference, making UAV swarm-based FFD missions particularly demanding [25]. To address these limitations, [26] develops a lightweight FFD model optimized for resource-constrained platforms like UAV swarm, enabling timely and accurate detection. Building on the critical importance of early FFD for damage mitigation, [6] introduces an innovative approach that fuses visible and infrared imagery with a custom dataset, overcoming accuracy and real-time processing limitations of existing methods.

Recent advances include a UAV-assisted edge computing system for precise fire detection and segmentation [7], and a lightweight hierarchical AI framework that dynamically switches between Machine Learning (ML) and deep learning CNN models to balance detection accuracy and computational efficiency [27].

The growing body of research underscores the advantages of UAV swarm in FFD missions. Successful implementation typically requires ML capabilities [28], yet traditional centralized ML approaches prove impractical due to bandwidth and energy constraints that hinder large-scale data transmission. In contrast, gossip learning emerges as an ideal decentralized alternative, enabling efficient edge intelligence for the UAV swarm [29].

C. Gossip Learning Innovations

Gossip Learning (GL) is a decentralized variant of Federated Learning (FL), and has been widely adopted in various applications. For instance, [13] examines how UAV inaccessibility impacts GL convergence in hostile environments, theoretically analyzing the effects of: the number of inaccessible UAVs, data non-IIDness, and duration of inaccessibility. This decentralized approach significantly enhances the training efficiency while optimizing the battery utilization of UAV swarm. Furthermore, [30] establishes the first performance baseline for GL on random time-varying graphs, characterizing its fundamental performance patterns in hostile mission scenarios and demonstrating its suitability for UAV swarm applications.

Unlike conventional FL that depends on centralized servers, which could introduce privacy risks, communication bottlenecks, and single-point failures, some recent advances have

developed improved decentralized solutions: [31] presents a privacy-enhanced, gossip-based FL method incorporating local differential privacy with asynchronous, hierarchical gradient aggregation through a pairwise gossip algorithm; [32] introduces GossipFL, a communication-efficient framework where a model sparsification algorithm reduces inter-client communication overhead, and an innovative gossip matrix generation algorithm optimizes bandwidth utilization while maintaining convergence guarantee.

For the hostile environments requiring continuous online learning, [33] develops Floating Gossip, a fully distributed GL scheme enabling opportunistic, privacy-preserving cooperative model training. [34] proposes a D2D-based GL approach that can improve the model instance of each node through transferring knowledge among nodes and on different strategies for estimating the potential contribution of neighboring nodes to the training process at one node.

These studies collectively demonstrate the strengths of GL in efficient model propagation and hostile environment adaptation. When implemented in UAV swarm for FFD missions, GL can offer three key advantages: (i) Enhanced mission efficiency through decentralized coordination; (ii) Improved scalability and robustness against UAV failures; (iii) Significant reduction in training overhead.

D. UAV Swarm Robustness

UAV swarm represents a dynamic form of the Internet of Things that is particularly vulnerable to partial unit destruction from environmental interference [35]. This inherent vulnerability has made UAV swarm robustness a critical research focus in the field.

While UAV swarm excels at self-organized and adaptive mission execution, their operational robustness presents significant challenges that demand specialized investigation. Previous research has developed several innovative approaches: [36] employs a semi-Markov process to model the state probability distribution of information exchange links as the UAV swarm exhibits the robustness behavior, proposing a resilience-based topology reconstruction method for optimal UAV replacement. [37] introduces an online Graph Convolutional Network (GCN) solution, which achieves real-time topology recovery of UAV swarm, and a self-healing trajectory planning algorithm is designed for communication restoration. [38] enhances security by formulating malicious UAV detection as a localization feasibility problem using position and distance data. Moreover, [39] draws inspiration from biological systems to design a failure-resistant swarm architecture. Likewise, [40] develops metaheuristic algorithms for collision-free, time-efficient path planning in disaster scenarios. Additionally, [41] establishes a comprehensive three-layer metric framework for the evaluation of UAV swarm robustness.

To address the issue of the UAV incapacitation, our work enhances the UAV swarm robustness through proposing RGLF which includes three fault-tolerant methods (GC, GB, and GS): GC method organizes UAVs into dispersed clusters, thus avoiding the complete loss of local data of each cluster; GB method

TABLE II
MAIN NOTATIONS

Parameter	Description
$\mathcal{N}$	UAV swarm
ψ	Number of clusters in $\mathcal{N}$
N_i	Number of incapacitated UAVs
N_a	Number of available UAVs
C_k	The k -th cluster in $\mathcal{N}$
$D_{loc}^{(t)}(v_i)$	Local dataset of UAV v_i at the t -th time slot
t^*	Gossip epoch (number of time slots)
$B^{(t)}(v_i)$	Business data collected from the visual coverage of UAV v_i at the t -th time slot
$D^{(t)}(C_k)$	Dataset of cluster C_k at the t -th time slot
V_k	Set of UAVs in cluster C_k
V'_k	Set of available UAVs in cluster C_k
$D(C_k, V'_k)$	Dataset of available UAVs V'_k in cluster C_k
$D(C_k, V_k \setminus V'_k)$	Dataset of incapacitated UAVs $V_k \setminus V'_k$ in cluster C_k
$\hat{D}(C_k, V'_k, V_k)$	Restored data of incapacitated UAVs $V_k \setminus V'_k$ through data backup stored by available UAVs V'_k in cluster C_k

is integrated into each cluster to enable the recovery of local data from incapacitated UAVs during FFD missions; GS method maintains the FFD mission continuity when one or more gossipmongers become incapacitated. Additionally, note that the training efficiency of UAV swarm in FFD missions can also be improved by the GL protocol due to the decentralized learning.

III. SYSTEM MODEL AND PROBLEM FORMULATION

Assuming that there are N UAVs in a UAV swarm, denoted by $\mathcal{N} = \{v_1, \dots, v_N\}$. The UAV swarm is mission-oriented, primarily responsible for distributed visual coverage and data collection for FFD missions. Communication and collaborative learning are auxiliary system functions that support the adaptability and robustness, rather than serving as the primary purpose of swarm formation. The following definitions are established to characterize the problem of UAV swarm robustness against the incapacitation of some UAVs. Table II shows the list of main notations.

A. UAV Swarm

Each UAV has the same communication range (denoted by R_c). R_c is typically large, e.g., the communication range of UAVs reaches 400 m [42], thus we assume that UAVs in the swarm can directly communicate with others. (e.g., the communication range of UAVs reaches 400 m [42], and the maximum communication range of a UAV even reaches 200 km [43]), thus we assume that UAVs in the UAV swarm can directly communicate with others. Although UAVs in the same cluster may be geographically separated, the communication range of UAVs is sufficiently large to support stable communications in FFD missions. In addition, proximity constraints and local clustering adjustments can be incorporated to further ensure communication stability when necessary. Once the UAV swarm arrives the airspace of the designated FFD missions, each UAV

initiates the data collection. The local dataset of v_i at the t -th time slot is denoted by $D_{loc}^{(t)}(v_i)$, which is composed of two parts: (i) The local model parameters of v_i , denoted by $w_i^{(t)}$; (ii) The local business data (videos and images) collected from the visual coverage of v_i , denoted by $D_{bus}^{(t)}(v_i) = \{B^{(0)}(v_i), \dots, B^{(t)}(v_i)\}$.

A cluster is a group of UAVs organized to support the coordination and communication during FFD missions, including data backup, fault-tolerant learning, and gossip operations. For a cluster C_k , UAVs in C_k apply a GB method to backup the local data collaboratively, and a shared cluster dataset $D^{(t)}(C_k)$ is constructed as:

$$D^{(t)}(C_k) = \bigcup_{v_i \in C_k} \{w_i^{(t)} \cup D_{bus}^{(t)}(v_i)\}. \quad (1)$$

Accordingly, the local dataset of UAV $v_i \in C_k$ at the t -th time slot includes its own model parameters and business data, as well as the data of other UAVs (in the same cluster) stored by v_i . Hence, $D_{loc}^{(t)}(v_i)$ is expressed as:

$$D_{loc}^{(t)}(v_i) = w_i^{(t)} \cup D_{bus}^{(t)}(v_i) \cup \bigcup_{v_j \in C_k \setminus v_i} \{w_j^{(t)} \cup D_{bus}^{(t)}(v_j)\}, \quad (2)$$

which can be compactly rewritten using the cluster dataset as:

$$D_{loc}^{(t)}(v_i) = D^{(t)}(C_k), \quad \text{for all } v_i \in C_k. \quad (3)$$

B. Objective Functions

The problem objectives are to maintain the mission performance under the incapacitation of some UAVs and reduce the data loss from the incapacitated UAVs. Assuming that there are N_i incapacitated UAVs and N_a available UAVs, where $N_i + N_a = N$. Let V'_k denote the set of available UAVs in the cluster C_k , where $N_a = \sum_{k=1}^{\psi} |V'_k|$. Note that the local datasets of the incapacitated UAVs can be restored through GB method. To assess the robustness of UAV swarm in FFD missions, the problem objectives are formally presented as follows:

$$\begin{cases} \max \frac{Acc'}{Acc}, \\ \min \sum_{k=1}^{\psi} \left(|V_k \setminus V'_k| - \frac{N_i}{\psi} \right)^2, \\ \min \sum_{k=1}^{\psi} \left| D(C_k, V_k \setminus V'_k) \setminus \widehat{D}(C_k, V'_k, V_k) \right|, \end{cases} \quad (4)$$

where Acc' and Acc denote the detection accuracy of FFD missions with and without the incapacitation of N_i UAVs, respectively. The ratio $\frac{Acc'}{Acc}$ serves as an indicator of performance maintenance under the UAV incapacitation. $(|V_k \setminus V'_k| - \frac{N_i}{\psi})^2$ denotes the deviation of the number of incapacitated UAVs in cluster C_k from $\frac{N_i}{\psi}$ (i.e. the average number of incapacitated UAVs per cluster). Since the GC method facilitates the dispersed distribution of UAVs, it reduces the risk of all UAVs being incapacitated in a cluster. $\sum_{k=1}^{\psi} |D(C_k, V_k \setminus V'_k) \setminus \widehat{D}(C_k, V'_k, V_k)|$ denotes the total size of lost data (due to the UAV incapacitation), where $D(C_k, V_k \setminus V'_k)$ and $\widehat{D}(C_k, V'_k, V_k)$ denote the dataset of incapacitated UAVs in the cluster C_k and the restored data of incapacitated UAVs through the data backup stored by available UAVs V'_k in the cluster C_k , respectively.

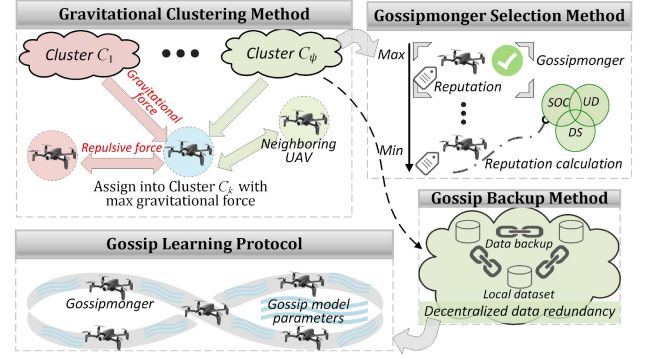


Fig. 3. Main components in RGLF.

IV. ROBUST GOSSIP LEARNING FRAMEWORK

In our proposed RGLF, the UAV swarm is first divided into several clusters by GC method. GS method enables each cluster selecting a representative UAV as the gossipmonger to participate in gossip learning protocol, which combines the collaborative training of the global model with the fault-tolerant model updating for the UAV swarm-based FFD missions. Besides, UAVs in each cluster upload their local datasets to the gossipmonger to form the cluster dataset by the GB method. Fig. 3 presents the main components in our proposed RGLF.

A. Gravitational Clustering Method

To enhance the robustness of UAV swarm against the incapacitation of some UAVs, GC method is first introduced for organizing UAVs into dispersed clusters. The details of GC method is given as follows:

In the UAV swarm $\mathcal{N}$, supposing that there are ψ clusters, and the set of clusters is denoted by $\mathcal{C}$. For a UAV v_i , the set of neighboring UAVs is denoted by $\mathcal{N}_n(v_i)$ and $\mathcal{N}_n(v_i) = \{v_j \in \mathcal{N} \mid L_2(v_i, v_j) \leq d_{max}\}$, where d_{max} represents the maximum distance for identifying neighboring UAVs, and $L_2(v_i, v_j)$ denotes the Euclidean distance between v_i and v_j .

In GC method, all UAVs are initially unclustered, and each cluster exerts an equal gravitational force on all unclustered UAVs. Notably, the gravitational force exerted by a cluster decreases as the size of the cluster increases. For instance, the gravitational force exerted by a cluster C_k on the unclustered UAVs is given by:

$$F_{gra}(C_k) = f_{gra} \cdot \left(\frac{N}{\psi} - |C_k| \right), \quad (5)$$

where f_{gra} and $|C_k|$ denote the initial gravitational force and the size of cluster C_k , respectively. By GC method, each UAV will join a cluster that is different from the clusters of its neighboring UAVs, ensuring that UAVs from the same cluster are evenly distributed in a dispersed manner to prevent the spatial aggregation. In addition, the size of each cluster (i.e., the number of UAVs in each cluster) is approximately balanced to enhance the UAV swarm robustness.

For v_i , only the neighboring UAVs in $\mathcal{N}_n(v_i)$ that have already been assigned into a cluster exert the repulsive forces on v_i .

Specifically, the repulsive force exerted on UAV v_i with respect to a cluster C_k is defined as:

$$F_{rep}(v_i, C_k) = \sum_{v_j \in \mathcal{N}_n(v_i)} \mathbb{I}(v_j \in C_k) \cdot \frac{f_{rep}}{L_2(v_i, v_j)}, \quad (6)$$

where $\mathbb{I}(v_j \in C_k)$ denotes an indicator function that outputs the value 1 when $v_j \in C_k$, and 0 otherwise. f_{rep} denotes the unit repulsive force. These repulsive forces from UAVs in $\mathcal{N}_n(v_i)$ reduce the gravitational forces of their respective clusters on v_i , thereby encouraging dispersed distribution of UAVs in the same cluster.

By combining these repulsive forces and gravitational forces, UAV v_i is assigned into the cluster C_k^* that maximizes the remaining gravitational force:

$$C_k^* = \arg \max_{C_k} (F_{gra}(C_k) - F_{rep}(v_i, C_k)). \quad (7)$$

In the case of a tie, UAV v_i is assigned into the cluster with the smallest index. The gravitational force $F_{gra}(C_k)$ that the cluster C_k exerts on other UAVs is reduced due to the increase of $|C_k|$, thereby promoting a balanced size of clusters. The above process continues until each UAV has been assigned into a cluster.

Note that UAVs in the same clusters are spatially dispersed. Such dispersed distribution of UAVs reduces the risk of all UAVs being incapacitated in a cluster, thereby alleviating the performance degradation of FFD missions. Moreover, the GC method guarantees that the size of each cluster is approximately balanced even after the incapacitation of some UAVs. If extreme cases occur (e.g., most of the UAVs have been incapacitated), some clusters may severely suffer from the UAV incapacitation, and the cluster reformation or merging must be conducted.

B. Gossip Backup Method

To enhance the robustness of UAV swarm in FFD missions, we propose the GB method for decentralized data redundancy. In each cluster, UAVs share and backup the local datasets to form a cluster dataset. This method ensures that even if one or more UAVs are incapacitated, the local datasets of incapacitated UAVs could be restored through the data backup of available UAVs in the same cluster. Essentially, the GB method ensures the dataset redundancy, and thus avoids the performance decline of RGLF caused by the incapacitation of some UAVs.

Importantly, when the incapacitated UAVs are not selected as gossipmongers, the GB method ensures that the detection accuracy of FFD missions will not be affected. Even in the cases where gossipmongers are compromised, the detection accuracy could be slightly affected in the current gossip epoch, because the gossipmongers are periodically reselected by the GS method.

C. Gossipmonger Selection Method

Fig. 4 illustrates the details of GS method. To avoid the failures of FFD missions due to the exits (incapacitation) of gossipmongers, the GS method is provided to dynamically select representative UAVs as gossipmongers for the collaborative training of the global model. Inspired by the dynamic server mechanism in [44], each cluster periodically selects one or more

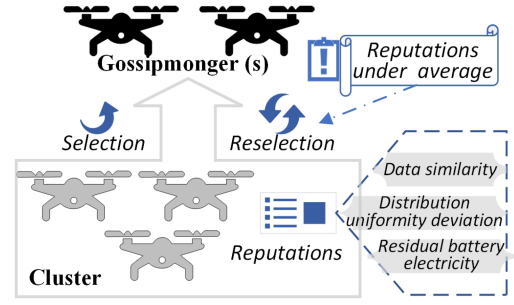


Fig. 4. Illustration of GS method.

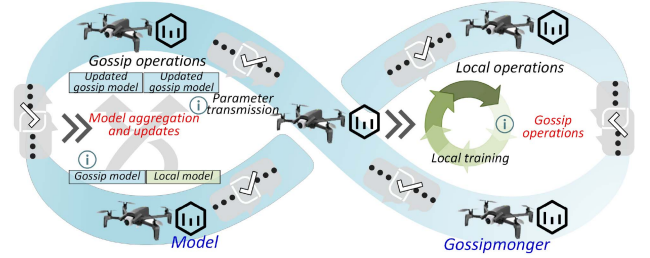


Fig. 5. Details of GL Protocol.

UAVs with high reputations to become gossipmongers, which considers some factors (e.g., residual battery electricity, data similarity, and distribution uniformity deviation).

Each cluster typically selects a small number of gossipmongers (e.g., one to four). Instead of fixed selections, gossipmongers are periodically reselected to mitigate the risk of performance degradation or unexpected UAV incapacitation. Specifically, every gossip epoch, each UAV calculates and sends its reputation to other UAVs in the same cluster. If the reputation of the current gossipmongers are lower than the average reputation of all UAVs in the same cluster, the UAV with the highest reputation is selected as the new gossipmonger. The new gossipmonger then informs the other UAVs about the update.

Moreover, to enhance the robustness and fairness, the current gossipmonger is excluded from the candidate set in the next selection whenever alternative qualified UAVs are available, preventing consecutive gossipmonger selection except in unavoidable cases, such as when the cluster size is very small.

D. Gossip Learning Protocol

We formalize the procedures of GL protocol. This formalization begins with defining the processes that take place at the gossipmongers (UAV selected by GS method), including local training and model updates. Then, we define the collaborative training of the global model. Finally, we outline the training dynamics of the GL protocol in the UAV swarm. Fig. 5 illustrates the details of GL protocol.

1) *Local Operations*: During the model training for FFD missions, the UAV swarm aims to determine an optimal mapping function $\mathcal{H}_w: \mathcal{X} \rightarrow \mathcal{Y}$, where $\mathcal{X}$ denotes the set of training samples, $\mathcal{Y}$ denotes the corresponding ground-truth labels, and w denotes the model parameters. The optimal parameters w^*

can be derived by minimizing the output of the samplewise loss function $l(\mathcal{H}_{\mathbf{w}}(\mathcal{X}), \mathcal{Y})$. This process is expressed as:

$$\mathbf{w}^* = \arg \min_{\mathbf{w}} F(\mathbf{w}) = \arg \min_{\mathbf{w}} \frac{\sum_{x \in \mathcal{X}, y \in \mathcal{Y}} l(\mathcal{H}_{\mathbf{w}}(x), y)}{|\mathcal{X}|}. \quad (8)$$

In GL protocol, each gossipmonger performs the gradient descent to train a local model. In particular, we adopt the Adaptive Moment Estimation (Adam) optimizer due to its computational efficiency and simplicity. Adam is well suited for the missions with many model parameters, and effective for handling non-stationary objectives. Each gossipmonger updates its local model as follows:

$$\mathbf{w}^{(t+1)} = \mathbf{w}^{(t)} - \eta \cdot \frac{\hat{m}_1^{(t+1)}}{\sqrt{\hat{m}_2^{(t+1)} + \epsilon}}, \quad (9)$$

where $\hat{m}_1^{(t+1)}$ and $\hat{m}_2^{(t+1)}$ denote the bias-corrected estimates of first moment estimate and the second moment estimate, respectively. η denotes the learning rate, and ϵ denotes a small constant.

2) *Gossip Operations*: During the k -th gossip epoch (e.g. at the t -th time slot), once a gossipmonger v_i receives the gossip model parameters $\tilde{\mathbf{w}}_k^{(t)}$ from another peer gossipmonger (which has already conducted the gossip operations), it performs the model aggregation by merging $\tilde{\mathbf{w}}_k^{(t)}$ with its local model $\mathbf{w}_i^{(t)}$. The aggregated model parameters $\tilde{\mathbf{w}}_{i,k}^{(t)}$ are computed using the FedAvg algorithm as follows:

$$\tilde{\mathbf{w}}_{i,k}^{(t)} = \alpha \cdot \mathbf{w}_i^{(t)} + (1 - \alpha) \cdot \tilde{\mathbf{w}}_k^{(t)}, \quad (10)$$

where $\alpha \in (0, 1)$ is a tunable coefficient that controls the contribution of $\mathbf{w}_i^{(t)}$ and $\tilde{\mathbf{w}}_k^{(t)}$. Typically, $\alpha = 0.5$ is used for equal-weight averaging. Then, gossipmonger v_i conducts a single local training epoch to update the aggregated model parameters $\tilde{\mathbf{w}}_{i,k}^{(t)}$, resulting in the updated gossip model parameters $\mathbf{w}_k^{(t+1)}$.

The updated gossip model parameters $\mathbf{w}_k^{(t+1)}$ are subsequently transmitted to the other peer gossipmonger (which has not yet conducted the gossip operations) for further gossip operations. This decentralized iterative process continues until all gossipmongers have participated in the current gossip epoch.

3) *Training Dynamics*: Initially, the gossip model is initialized with the same model parameters as the pre-trained model distributed to each UAV. The first gossip epoch serves as a warm-up phase to stabilize the gossip learning protocol, and this decentralized iterative process proceeds according to a randomly determined sequence of gossipmongers. Upon completing the gossip operations, each gossipmonger resumes the local training by executing a number of local epochs until it receives the updated gossip model parameters in the next gossip epoch. Consequently, the local training time for each gossipmonger during the warm-up phase varies depending on its position in the sequence. The later a gossipmonger appears in the sequence, the more local training time it performs before receiving the gossip model parameters. Accordingly, in the subsequent gossip epochs, all gossipmongers undergo an equal number of local epochs between gossip operations.

When a gossipmonger is replaced, the new gossipmonger inherits the model training progress of its predecessor. Upon completing its gossip operations, it releases the updated gossip model parameters to all UAVs in the UAV swarm to update their local models.

The GL protocol continues until a predefined convergence condition is satisfied. Specifically, the goal of GL protocol is to minimize that:

$$f(\mathbf{w}) := \frac{1}{N_g} \sum_{i=1}^{N_g} f_i(\mathbf{w}), \quad (11)$$

where N_g denotes the number of gossipmongers. To derive the optimal parameters $\mathbf{w}^*$, this goal can be rewritten as:

$$\mathbf{w}^* = \inf \sum_{i=1}^{N_g} f_i(\mathbf{w}^{(t)}), \quad (12)$$

where $f_i(\mathbf{w}^{(t)})$ denotes the loss function of gossipmonger v_i , which is μ -smooth, and convex. For any gossipmonger (e.g. v_i), there is:

$$\|\nabla f_i(\mathbf{w}) - \nabla f_i(\tilde{\mathbf{w}})\| \leq \mu \cdot \|\mathbf{w} - \tilde{\mathbf{w}}\|, \quad (13)$$

where $\nabla f_i(\mathbf{w})$ denotes the gradient of $f_i(\mathbf{w})$. Besides, the unbiased stochastic gradient $g_i(\mathbf{w})$ is expressed as:

$$\mathbb{E}[g_i(\mathbf{w}) \mid \mathbf{w}] = \nabla f_i(\mathbf{w}). \quad (14)$$

Then, the variance of these gradients is bounded by σ , i.e.,

$$\mathbb{E}[\|g_i(\mathbf{w}) - \nabla f_i(\mathbf{w})\|^2 \mid \mathbf{w}] \leq \sigma^2. \quad (15)$$

Moreover, the gradient divergence is bounded by τ , i.e.,

$$\|\nabla f_i(\mathbf{w}_i^{(t)}) - \nabla f(\mathbf{w}^{(t)})\|^2 \leq \tau^2. \quad (16)$$

Thus, we consider that the training ends in the k -th gossip epoch once the global model parameters $\mathbf{w}_k$ satisfy the convergence criterion, otherwise the next gossip epoch starts. The processes that take place during our designed GL protocol are summarized in Algorithm 1.

V. THEORETICAL ANALYSIS OF RGLF

A. Complexity

Table III shows the communication overhead and computational complexity of our proposed RGLF.

1) *Complexity of Gossip Clustering*: UAVs broadcast their current states (e.g. coordinates and cluster assignment), incurring a communication overhead of $O(N)$. Moreover, UAVs in the same cluster could exchange the local datasets with each other, leading to a communication overhead of $O(\frac{N^2}{\psi})$. Therefore, the communication overhead for the GC method is written as: $O(N + \frac{N^2}{\psi})$. In terms of the computations, each UAV evaluates the gravitational forces from ψ clusters, and the set of neighboring UAVs typically includes m UAVs. Thus, the computational complexity scales to $O(N \cdot (\psi + m))$.

TABLE III
OVERHEAD AND COMPLEXITY OF RGLF

Module	Communication overhead	Computational complexity
GC	$O(N + \frac{N^2}{\psi})$	$O(N \cdot (\psi + m))$
GS	$O(\frac{N}{\psi} + \psi^2)$	$O(N)$
GL	$O(k \cdot \mathbf{w})$	$O(L \cdot K_s^2 \cdot C_{in} \cdot C_{out} \cdot W \cdot H)$
Total	$O(N + \frac{N}{\psi} + \frac{N^2}{\psi} + \psi^2 + k \cdot \mathbf{w})$	$O(L \cdot K_s^2 \cdot C_{in} \cdot C_{out} \cdot W \cdot H + N \cdot \psi + N \cdot m + N)$

Algorithm 1: Gossip Learning (GL) Protocol.

```

1: Input: Initial model parameters  $\mathbf{w}^{(0)}$ , learning rate  $\eta$ ,
   weight coefficient  $\alpha$ , gossipmonger number  $N_g$ 
2: Initialize each gossipmonger  $v_i$  with  $\mathbf{w}_i^{(0)} = \mathbf{w}^{(0)}$ 
3: Randomly determine sequence of gossipmongers
    $\{v_1, v_2, \dots, v_{N_g}\}$ 
4: while every gossip epoch  $k$  do
5:   Initialize gossip model  $\tilde{\mathbf{w}}_k^{(t)}$ 
6:   for each gossipmonger  $v_i$  do
7:     perform local training  $\mathbf{w}_i^{(t)} \leftarrow \text{LocalTrain}(\mathbf{w}_i^{(t)})$ 
8:   end for
9:   for each gossipmonger  $v_i$  in sequence do
10:    // Other gossipmongers perform local training
11:    if  $i = 1$  then
12:       $\tilde{\mathbf{w}}_k^{(t)} \leftarrow \mathbf{w}_j^{(t)}$ 
13:    else
14:      Aggregate:  $\tilde{\mathbf{w}}_{i,k}^{(t)} \leftarrow \alpha \cdot \mathbf{w}_i^{(t)} + (1 - \alpha) \cdot \tilde{\mathbf{w}}_k^{(t)}$ 
15:      Training:  $\mathbf{w}_k^{(t+1)} \leftarrow \text{LocalTrain}(\tilde{\mathbf{w}}_{i,k}^{(t)})$ 
16:      Update:  $\tilde{\mathbf{w}}_k^{(t+1)} \leftarrow \mathbf{w}_k^{(t+1)}$ 
17:    end if
18:    // Resume local training
19:  end for
20:  if convergence criterion is satisfied then
21:    break
22:  end if
23:   $k \leftarrow k + 1$ 
24: end while
25: Output: Final global model  $\mathbf{w}_k$ 

```

2) *Complexity of Gossipmonger Selection:* Each UAV periodically transmits its reputation to other UAVs in the same cluster and receives the gossipmonger selection results. Since this process only involves intra-cluster transmission and selection, the communication overhead reaches $O(\psi \cdot (\frac{N}{\psi^2} + \psi))$, where $O(\frac{N}{\psi^2} + \psi)$ denotes the communication overhead in each cluster. With respect to the computations, the GS method includes calculating the reputations and performing comparisons in clusters, which incurs a computational complexity of $O(N)$.

3) *Complexity of Gossip Learning:* In GL, each gossip epoch involves the transmission of gossip model parameters $\mathbf{w}$ between gossipmongers. Over k gossip epochs, the communication overhead scales to $O(k \cdot |\mathbf{w}|)$, where $|\mathbf{w}|$ denotes the size of the gossip model parameters. Besides, the computational complexity for training the model is written as:

$$O(L \cdot K_s^2 \cdot C_{in} \cdot C_{out} \cdot W \cdot H + \xi \cdot W' \cdot H' \cdot C_{out}), \quad (17)$$

where $W \times H$ and $W' \times H'$ ($W' \times H' \ll W \times H$) denote the dimensions of the input images and the feature maps after passing through the convolutional layers, respectively. $K_s \times K_s$, L , and ξ denote the convolution kernel size, number of layers, and number of classes, respectively. C_{in} and C_{out} denote the number of input/output channels in each layer, respectively.

Thus, the total communication overhead of RGLF is of $O(N + \frac{N}{\psi} + \frac{N^2}{\psi} + \psi^2 + k \cdot |\mathbf{w}|)$, and the total computational complexity of RGLF is of $O(L \cdot K_s^2 \cdot C_{in} \cdot C_{out} \cdot W \cdot H + N \cdot \psi + N \cdot m + N)$.

B. Model Convergence

For the analysis of model convergence, we focus on the alternative deviation of local gradients and convergence upper bound of (11).

Proposition 1. When $N_g > 4$, for all $\mathbf{w}, t$,

$$\left\| \sum_{j=1}^{N_g} [\nabla f_i(\mathbf{w}_i^{(t)}) - \nabla f_j(\mathbf{w}_j^{(t)})] \right\|^2 \leq \frac{2N_g \cdot \tau^2}{N_g - 4}. \quad (18)$$

Proof. By adding and subtracting the item $\nabla f(\mathbf{w}^t)$, (18) can be written as:

$$\begin{aligned}
Z(\mathbf{w}, t) &= \left\| \sum_{j=1}^{N_g} [\nabla f_i(\mathbf{w}_i^{(t)}) - \nabla f_j(\mathbf{w}_j^{(t)})] \right\|^2 \\
&= \left\| \nabla f_i(\mathbf{w}_i^{(t)}) - \sum_{j=1}^{N_g} \nabla f_j(\mathbf{w}_j^{(t)}) \right\|^2 \\
&= \left\| \nabla f_i(\mathbf{w}_i^{(t)}) - \nabla f(\mathbf{w}^t) + \nabla f(\mathbf{w}^t) - \sum_{j=1}^{N_g} \nabla f_j(\mathbf{w}_j^{(t)}) \right\|^2 \\
&= \left\| \nabla f_i(\mathbf{w}_i^{(t)}) - \nabla f(\mathbf{w}^t) + \frac{1}{N_g} \sum_{i'=1}^{N_g} \nabla f_{i'}(\mathbf{w}_{i'}^{(t)}) \right. \\
&\quad \left. - \frac{1}{N_g} \sum_{i'=1}^{N_g} \sum_{j=1}^{N_g} \nabla f_j(\mathbf{w}_j^{(t)}) \right\|^2. \quad (19)
\end{aligned}$$

Considering that $(\|a + b\|^2) \leq 2(\|a\|^2 + \|b\|^2)$, and according to the definition of τ^2 , we obtain that:

$$\begin{aligned}
Z(\mathbf{w}, t) &\leq 2 \left\| \nabla f_i(\mathbf{w}_i^{(t)}) - \nabla f(\mathbf{w}^t) \right\|^2 \\
&\quad + 2 \left\| \frac{1}{N_g} \sum_{i'=1}^{N_g} \left[\nabla f_{i'}(\mathbf{w}_{i'}^{(t)}) - \sum_{j=1}^{N_g} \nabla f_j(\mathbf{w}_j^{(t)}) \right] \right\|^2 \\
&\leq 2\tau^2 + \frac{2}{N_g} \sum_{i'=1}^{N_g} \left\| \nabla f_{i'}(\mathbf{w}_{i'}^{(t)}) - \sum_{j=1}^{N_g} \nabla f_j(\mathbf{w}_j^{(t)}) \right\|^2. \quad (20)
\end{aligned}$$

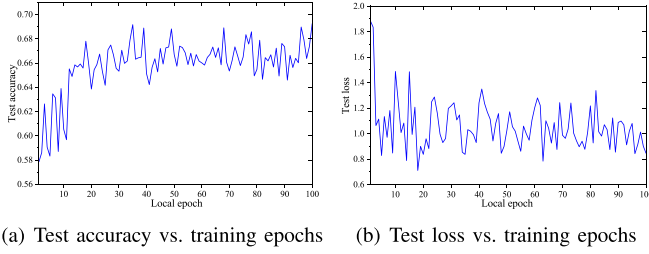


Fig. 6. Converge of RGLF.

Hence, there is:

$$\begin{aligned}
 & \left(1 - \frac{2}{N_g}\right) \left\| \sum_{j=1}^{N_g} [\nabla f_i(\mathbf{w}_i^{(t)}) - \nabla f_j(\mathbf{w}_j^{(t)})] \right\|^2 \\
 & \leq 2\tau^2 + \frac{2}{N_g} \sum_{i'=1, i' \neq i}^{N_g} \left\| \nabla f_{i'}(\mathbf{w}_{i'}^{(t)}) - \sum_{j=1}^{N_g} \nabla f_j(\mathbf{w}_j^{(t)}) \right\|^2 \\
 & \leq 2\tau^2 + \frac{2}{N_g} \sum_{i'=1}^{N_g} \left\| \nabla f_{i'}(\mathbf{w}_{i'}^{(t)}) - \sum_{j=1}^{N_g} \nabla f_j(\mathbf{w}_j^{(t)}) \right\|^2.
 \end{aligned} \tag{21}$$

By arranging the inequality, (21) is rewritten as:

$$\left(1 - \frac{4}{N_g}\right) \left\| \sum_{j=1}^{N_g} [\nabla f_i(\mathbf{w}_i^{(t)}) - \nabla f_j(\mathbf{w}_j^{(t)})] \right\|^2 \leq 2\tau^2. \tag{22}$$

□

Proposition 1 provides an upper bound on the deviation of local gradients, given by $\frac{2N_g\tau^2}{N_g-4}$ for $N_g > 4$, indicating that as the number of gossipmongers increases, the gradient divergence across gossipmongers is effectively reduced, contributing to more stable model update and converge. According to [45], we obtain the convergence upper bound on the expectation of $\mathbb{E}[\|\nabla f(\mathbf{w}^t)\|^2]$:

$$\begin{aligned}
 & \min_t \mathbb{E} [\|\nabla f(\mathbf{w}^t)\|^2] \\
 & \leq O \left(\frac{f(\mathbf{w}^0) - f(\mathbf{w}^*)}{B_s \cdot \eta} + \frac{\tau^2}{N_g - 4} + \eta^2 + N_g \cdot \tau^2 \right. \\
 & \quad \left. + B_s \cdot \mu^2 \cdot \eta^2 \cdot \sigma^2 + \frac{\mu \cdot \eta \cdot \rho^2 \cdot \sigma^2}{N_g} \right)
 \end{aligned} \tag{23}$$

where $O(\cdot)$ denotes the asymptotic upper bound (omitting constant factors), and $\eta \leq \frac{1}{8B_s \cdot \mu \cdot N_g}$, $N_g > 4$. The convergence bound in (23) highlights how increasing the number of gossipmongers (N_g) mitigates the impact of gradient divergence (τ^2), while properly tuning the learning rate η and batch size B_s ensures the stability and efficient convergence.

As presented in Fig. 6, the model convergence can be guaranteed in expectation as long as the parameters (N_g, η, B_s) are properly set. The X -axis denotes the training epochs, and

TABLE IV
SIMULATION PARAMETERS

Parameter	Description	Value
N	Number of UAVs	100
ψ	Number of clusters in $\mathcal{N}$	5
N_g	Number of gossipmongers	10
d_{max}	Maximum distance for establishing neighboring UAVs	80 m
R_c	Maximum communication range of UAVs	400 m
E_{init}	Initial battery energy of UAV	0.0222 kwh
t^*	Gossip epoch (number of time slots)	5
η	Learning rate	0.01
m_1	First moment estimate	0.9
m_2	Second moment estimate	0.999
w_d	Weight decay	0.0001
B_s	Batch size	64

the Y -axis denotes the test accuracy (Fig. 6(a)) and test loss (Fig. 6(b)), respectively. During the first 20 epochs, the mission performance of our proposed RGLF are rapidly improved. In the subsequent epochs, the test accuracy stabilizes around 0.67, and the test loss stabilizes around 1.0. These results also imply that RGLF can converge quickly.

VI. PERFORMANCE EVALUATIONS

In this section, we provide comprehensive performance evaluations for our proposed RGLF, along with the comparisons to several other training methods. To provide system-level validation, the evaluations are designed to jointly examine the mission robustness and effectiveness, training efficiency and stability, data redundancy, and scalability. These aspects are validated through comparative experiments, ablation studies, sensitivity analyses, and robustness evaluations, which collectively characterize the operational performance of RGLF in UAV swarm-based FFD missions.

The simulations are carried out using the FLAME dataset [46], which consists of the aerial data acquired by UAVs during the controlled burns of forest debris in an Arizona pine woodland. FLAME dataset includes the videos and thermal heatmaps collected via infrared cameras installed on UAVs. The fire images captured by UAVs typically contain visible flames, and the presence of flames serves as a key indicator for fire detection. Based on this dataset, we simulate the UAV swarm-based FFD missions. Key simulation parameters are outlined in Table IV.

In RGLF, we adopt ResNet18 as the model for UAV swarm-based FFD missions. As a lightweight model in the ResNet family, ResNet18 offers a favorable trade-off between the mission performance and computational overhead, making it suitable for the deployment on resource-constrained platforms (such as UAVs). Compared with other deeper models like ResNet34 and ResNet101, ResNet18 better meets the real-time applications. Some examples of the detection results on FLAME dataset are shown in Fig. 7.

A. Comparisons Among Different Training Methods

To analyze the merits of RGLF, we compare RGLF with centralized learning, local learning, FL, HFL, and Partial Federated Learning (PFL) in terms of detection accuracy, loss, and training time. RGLF is designed to tackle the challenges in UAV



Fig. 7. Examples of some FFD results.

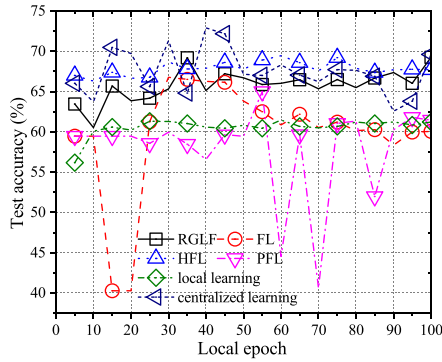


Fig. 8. Comparisons among different training methods (test accuracy).

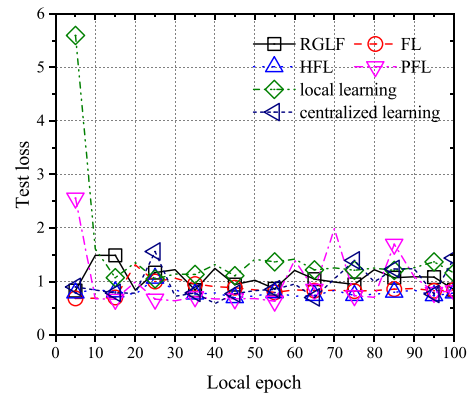


Fig. 9. Comparisons among different training methods (test loss).

swarm-based FFD missions, especially enhancing the robustness and reducing the training overhead through a fault-tolerant and decentralized learning framework.

In the centralized learning manner, all UAVs transmit their local datasets to a central server which trains the global model. In the local learning manner, each UAV trains the local model only on its local dataset, without any exchanges of data or model parameters with neighboring UAVs or the central server. FL enables UAVs to collaboratively train the global model by sharing local model updates with a central server, without transmitting their local datasets. In HFL, UAVs are assigned into clusters where local model parameters are first aggregated by the lower-tier servers in different clusters, and then the aggregated models are transmitted to the upper-tier server for a global model aggregation. Unlike FL, PFL takes a partial training-based approach that allows different parts of the global model to be trained. Our proposed RGLF adopts a decentralized gossip learning protocol combined with fault-tolerant model updating and some methods (GC, GB, and GS).

As shown in Fig. 8, the detection accuracy of forest fires obtained by the centralized learning, HFL, and RGLF is greater than that obtained by the local learning, FL, and PFL. However, it is important to note that the centralized learning is impractical for the UAV swarm due to the extremely high training overhead and poor robustness (if the central server is incapacitated, all FFD missions inevitably fail). Likewise, HFL relies on both lower-tier and upper-tier servers for the model parameter aggregation and updates, making HFL vulnerable to the incapacitation of lower-tier and upper-tier servers.

Furthermore, as illustrated in Fig. 9, RGLF yields much smaller loss compared with other methods due to the following mechanisms adopted in RGLF: (i) The decentralized gossip learning protocol enables gossipmongers to perform local model

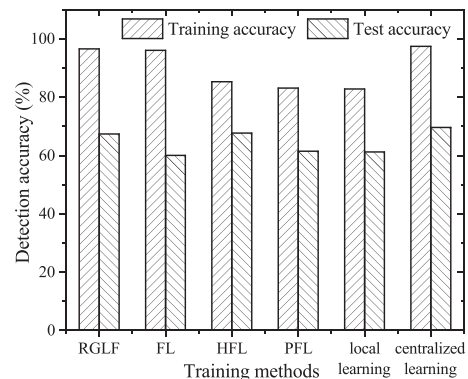


Fig. 10. Comparisons among different training methods (train accuracy and test accuracy).

updates and exchange model parameters with peers without relying on a central server, thereby maintaining the FFD mission continuity even in hostile mission environments. (ii) The utilization of the GB method effectively mitigates the impact of data heterogeneity among gossipmongers, which is critical for ensuring efficient convergence and improved performance.

Figs. 10 and 11 illustrate the comparisons among different methods in terms of detection accuracy and loss. RGLF outperforms the local learning, FL, and PFL in terms of detection accuracy and loss, because these training methods (local learning, FL, and PFL) typically suffer from the client drift¹ which is due to the fact that the model training relies on the local datasets, and the client drift inevitably decelerates the

¹A drift in the updates of each client results in slow and unstable convergence [47].

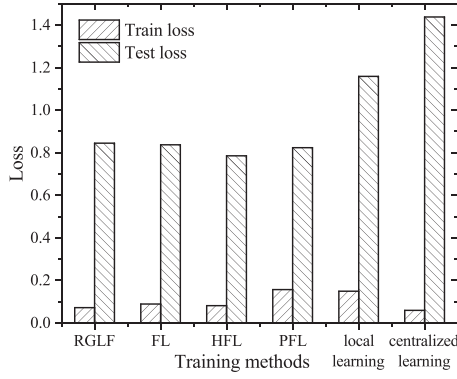


Fig. 11. Comparisons among different training methods (train loss and test loss).

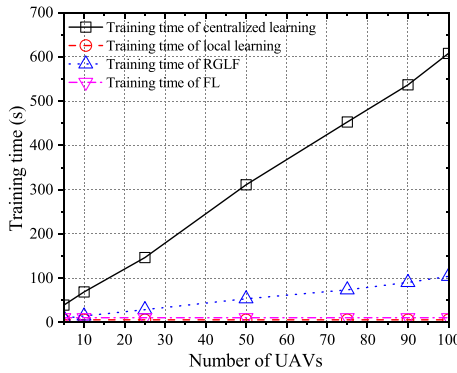


Fig. 12. Comparisons among different training methods on training time.

model convergence and reduces the detection accuracy. However, RGLF mitigates the impact of client drift through the robustness operations that expands the local datasets to form the cluster datasets.

Despite producing the smallest training overhead, the local learning yields the lowest detection accuracy among these training methods, indicating its inherent limitations for the UAV swarm-based FFD missions. In addition, the performance of HFL is marginally higher than that of FL, since HFL aggregates the model parameters in a two-tier learning manner for improving the detection accuracy, while FL only aggregates the model parameters for once. Moreover, RGLF exhibits a distinct advantage in terms of detection accuracy (train accuracy and test accuracy) and loss (train loss and test loss).

The training time consists of both communication delay and computation delay. In these training methods, centralized learning and local learning do not require model parameter aggregation. In contrast, both FL and PFL perform model parameter aggregation at each global epoch. Additionally, HFL involves two tiers of model parameter aggregation, but its training time remains comparable to that of FL. In our proposed RGLF, model parameters are periodically disseminated to other UAVs. Therefore, we only compare the training time among centralized learning, local learning, FL, and RGLF, and the results are provided in Fig. 12.

As illustrated in Fig. 12, we compare the training time of different training methods over five local epochs (i.e. one global

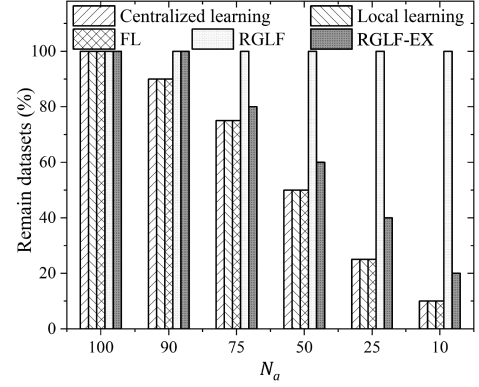


Fig. 13. Remaining datasets among different training methods under different N_a .

epoch). FL and local learning obtain the similar training time. In contrast, the training time of RGLF and centralized learning increases as the number of UAVs increases. This is due to the utilization of GB method, where UAVs in the same cluster are required to upload their local datasets to form the cluster datasets, thus leading to the increased communication delay in RGLF. In addition, as the number of UAVs increases, the cluster datasets are enlarged, thereby increasing the computational delay of RGLF.

Moreover, centralized learning suffers from a significantly higher training time compared with other methods, due to the need of aggregating and processing the local datasets of all UAVs, which underscores its impracticality for the UAV swarm-based FFD missions. Compared with centralized learning manner, RGLF maintains a relatively acceptable training time, making it more feasible in such situations.

These above simulation results show that RGLF can make a preferable trade-off between the detection accuracy and training time. Moreover, and these results also highlight its adaptability in the UAV swarm-based FFD missions.

B. Effect of Gossip Backup Method

To enhance the robustness of UAV swarm-based FFD missions, the GB method is employed, providing the redundancy and restoration for the local datasets of incapacitated UAVs.

As shown in Fig. 13, the remaining datasets under the UAV incapacitation are compared across different training methods. RGLF-EX (RGLF under extreme risks of UAV incapacitation) is included to provide a more intuitive illustration of the simulation results. In this extreme scenario, UAV incapacitation occurs in the worst case, i.e., once a cluster is incapacitated (all UAVs in the cluster become incapacitated), and the clusters are sequentially incapacitated. The simulation results show that with the GB method, RGLF can retain as much data as possible by restoring the local datasets through the data backup of incapacitated UAVs. Meanwhile, with the decrease of N_a , the remaining datasets are sharply reduced with FL, centralized learning, and local learning. Note that RGLF-EX also retains more of the remaining datasets compared with FL, centralized learning, and local learning, although the remaining datasets with RGLF-EX

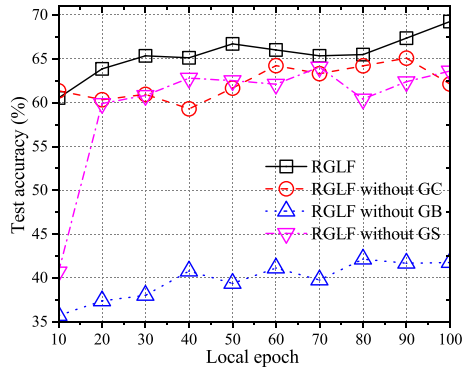


Fig. 14. Ablation results on test accuracy.

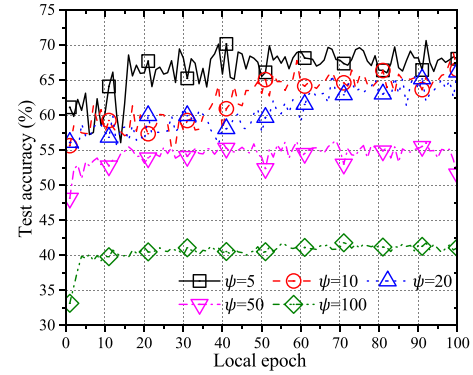
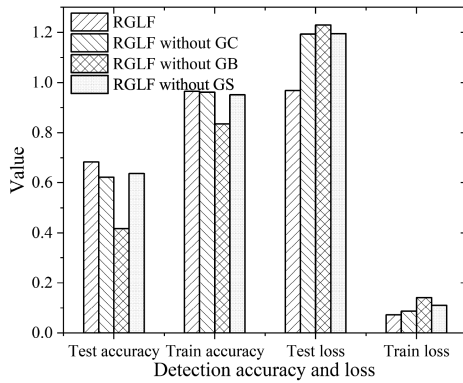
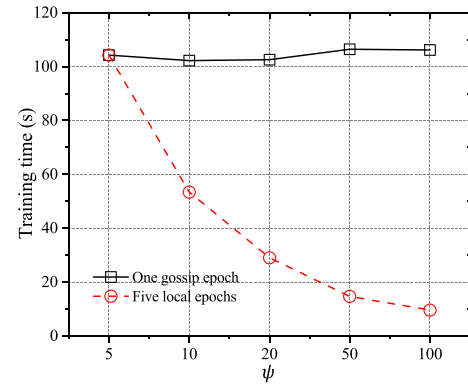
Fig. 16. Impact of ψ on test accuracy.

Fig. 15. Ablation results.

Fig. 17. Impact of ψ on training time.

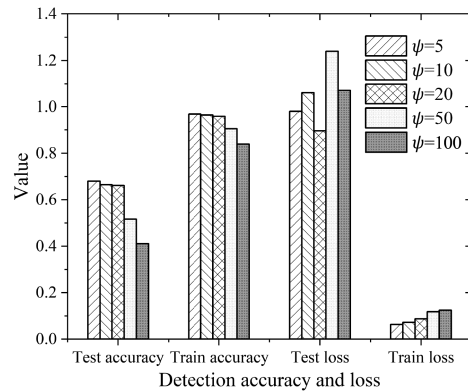
are also reduced. Therefore, our proposed RGLF can ensure the dataset redundancy, and thus avoid the performance decline caused by the incapacitation of some UAVs.

We further conduct ablation studies to validate the effects of GC, GB, and GS, respectively. As shown in Figs. 14 and 15, the complete RGLF model achieves the highest detection accuracy and the lowest loss. When GC is removed, the detection accuracy drops slightly, and losses increase moderately. In contrast, removing GB causes a significant degradation in mission performance, indicating that GB is particularly critical for model generalization. The absence of GS also leads to a noticeable performance drop, but less severe than GB. In summary, these results indicate that each of the three components exerts a unique and indispensable effect on the mission performance.

C. Impact of Cluster Number

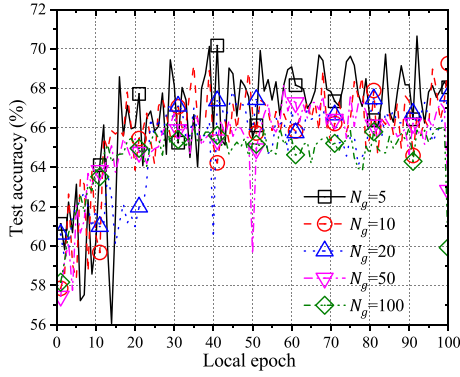
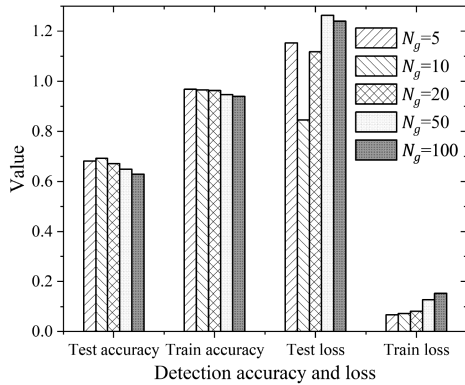
To investigate the impact of the number of clusters ψ , we first partition the test set into several subsets, enabling each UAV in the cluster to assess its mission performance on the corresponding test subset. Each UAV cluster selects a representative UAV as the gossipmonger to participate in the collaborative training of the global model.

As illustrated in Fig. 16, the variations in ψ significantly affect the mission performance. Specifically, a decrease in ψ gives rise to an increased cluster size and a reduced heterogeneity among different cluster datasets, which consequently leads to the

Fig. 18. Impact of ψ .

improved detection accuracy. Meanwhile, as shown in Fig. 17, under the condition of five local epochs, the training time decreases with the increase of ψ because fewer UAVs upload their local datasets to form the cluster datasets, and thus the training time in each cluster is shortened. Although the training time for one local epoch decreases as ψ increases, a gossip epoch consists of ψ local epochs, so the total training time for one gossip epoch remains approximately the same across different value of ψ .

Fig. 18 gives the results of the UAV swarm-based FFD missions under different cluster number ψ . When the cluster number ψ is set to 5, 10, and 20, RGLF achieves favorable detection

Fig. 19. Impact of N_g on test accuracy.Fig. 20. Impact of N_g .

accuracy and loss. In the following simulations, we adopt the setting $\psi = 5$ and explore the impact of increasing the number of gossipmongers in the global gossip process.

D. Impact of Gossipmonger Number

In Figs. 19 and 20, we investigate the impact of varying the gossipmonger number N_g . As depicted in Fig. 19, although the test accuracy maintains similar across different value of N_g , a subtle downward trend of the test accuracy is observed when N_g increases. These observations are attributed to the extended gossip epochs caused by a larger number of gossipmongers (each gossipmonger receives the aggregated gossip model parameters less frequently), which results in prolonged local adaptation. Consequently, the risk of local overfitting [48] increases, making RGLF hard to yield the optimal outcomes. When N_g is set to 50 and 100, the local overfitting problem appears, whereas it is observed only once when $N_g = 20$. These results highlight that selecting the representative UAVs as gossipmongers to participate in the collaborative training of the global model, rather than all UAVs in the UAV swarm, helps mitigate the local overfitting problem and achieves higher detection accuracy of the UAV swarm-based FFD missions.

In addition, Fig. 20 illustrates the detection accuracy and loss by varying the value of N_g . When N_g is set to 5, 10, and 20, RGLF achieves preferable mission performance. Notably, the configuration with $N_g = 10$ achieves the highest test accuracy and the lowest test loss, indicating that a moderate number

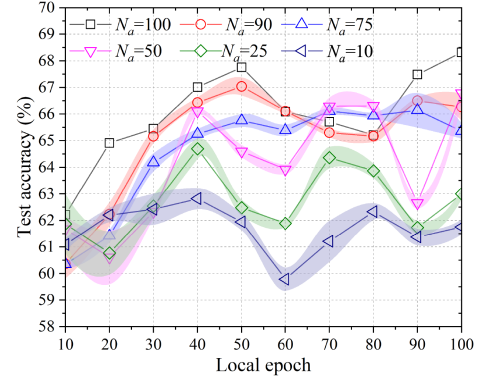
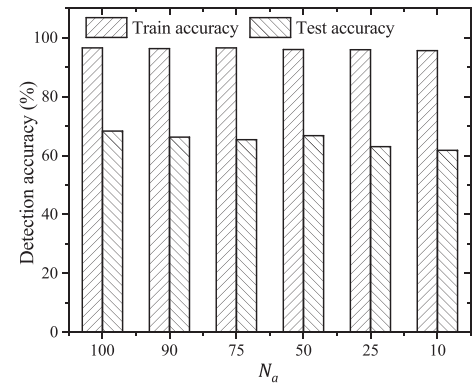
Fig. 21. Test accuracy vs. N_a .

Fig. 22. Robustness of FFD missions in terms of detection accuracy.

of gossipmongers in the global gossip process can improve the mission performance. These observations are attributed to the following facts: A properly chosen N_g balances the local adaptation and global gossip, mitigating the local overfitting problem caused by the infrequent gossip (i.e. long-term gossip epochs) while reducing the training overhead of gossipmonger associated with frequent gossip (i.e. short-term gossip epochs). Based on these findings, N_g is set to 10 in the following simulations.

E. Robustness of FFD Missions

To evaluate the robustness of FFD missions against the UAV incapacitation, a series of simulations are conducted (Figs. 21, 22, 23, and 24). As shown in Fig. 21, the detection accuracy decreases as N_a decreases due to the diminished availability of UAVs for participating in FFD missions. Nevertheless, RGLF introduces three fault-tolerant methods (GC, GB, and GS) that effectively alleviate the mission performance degradation caused by the UAV incapacitation.

Additionally, Figs. 22 and 23 further demonstrate that the performance of FFD missions obtained by RGLF remains relatively stable as N_a decreases (the increase of N_i). With a reduction in the number of available UAVs, the detection accuracy is decreased, and the loss is increased. This is because as N_a decreases, maintaining fault-tolerant model updating during the model training becomes challenging, which slows down the model convergence and undermines the effectiveness of FFD

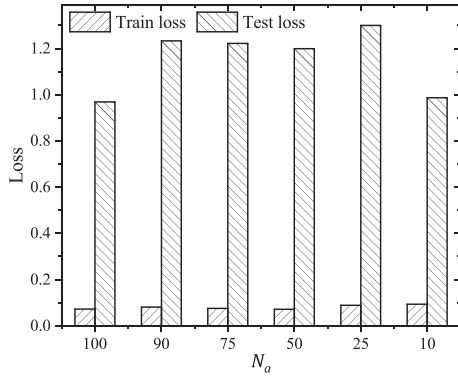


Fig. 23. Robustness of FFD missions in terms of loss.

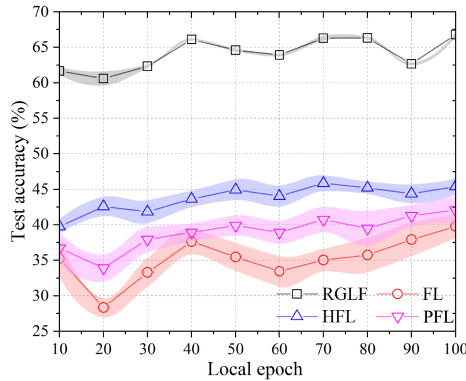


Fig. 24. Robustness of FFD missions among different methods.

missions. In such UAV swarm-based FFD missions, RGLF plays a key role in maintaining the mission performance despite the incapacitation of some UAVs.

Fig. 24 illustrates the impact of UAV incapacitation on mission performance, where half of the UAVs become incapacitated. It is evident that FL, HFL, and PFL suffer from notable fluctuations and degraded test accuracy. Specifically, FL exhibits the most pronounced instability, while HFL and PFL show moderate improvements but still remain vulnerable to data loss caused by the UAV incapacitation. In stark contrast, RGLF maintains a high and stable test accuracy, consistently staying above 60%. The utilization of the GB method allows RGLF to effectively mitigate the negative impacts of UAV incapacitation, ensuring efficient convergence and improved performance. These results confirm that RGLF is particularly well-suited for the deployment in hostile environments where the availability of UAVs cannot be guaranteed.

Overall, the collective evidence from the simulations corroborates the effectiveness of RGLF from various system-level perspectives. These complementary evaluations, taken together, substantiate the robustness of the proposed RGLF for UAV swarm-based FFD missions.

VII. CONCLUSION

In this work, we have proposed a Robust Gossip Learning Framework (RGLF) to improve the robustness and adaptability

of UAV swarm-based FFD missions in hostile mission environments. RGLF adopts a decentralized gossip learning protocol to enable fault-tolerant model updating without the need for a central coordinator, enhancing the adaptability of UAV swarm-based FFD missions. To further enhance the robustness of UAV swarm against the incapacitation of some UAVs, RGLF proposes a gravitational clustering method, a gossip backup method, and a gossipmonger selection method, thus enhancing the robustness of UAV swarm-based FFD missions.

Although the mission performance of RGLF has been verified by extensive simulations, several issues need further exploration in the future: (i) By leveraging solar, wind, or RF energy harvesting, UAVs can potentially sustain FFD missions longer, improving the adaptability of FFD missions. (ii) In some scenarios where UAVs are highly sparsely distributed, a gossip learning protocol with sparsified and adaptive communication could be considered to perform the FFD missions more effectively.

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